usage practices, traceability systems, hygiene principles and sanitary practices followed at the farm level by the operators

So what benefits does this scheme offer:

- reduce the risk of microbial contamination in fruits, vegetables and nuts and aims to make sure these foods are safe for you to eat.
- Enables producers in making their systems more sustainable through improved system implementation.
- Visibility of the certified producers on the global database giving access to local and global customers, markets, suppliers and retailers.
- Gives buyers a strong sense of trust based on the assurance of having a certified system in place.

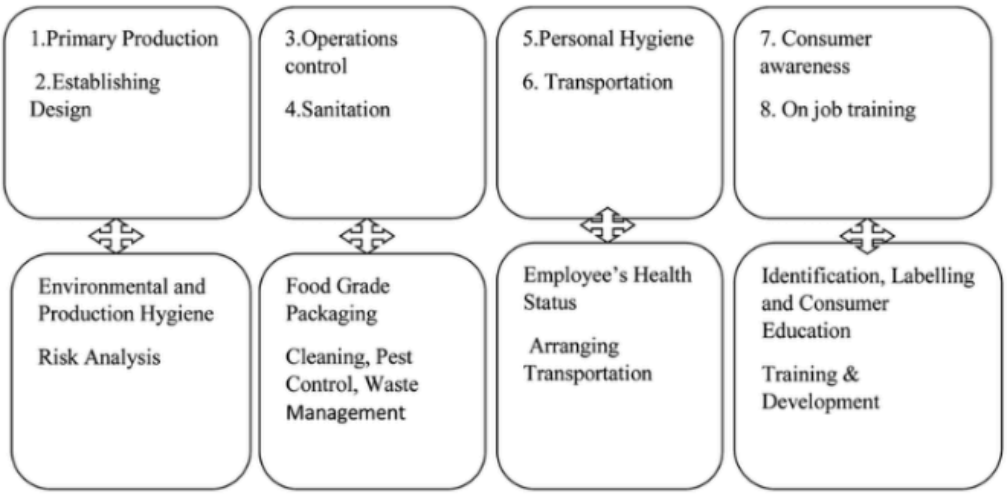
If you were to become GAP certified, this opens new market opportunities for your operation. Not only will you have access to new markets, you will also be reducing the risk of foodborne illnesses, and therefore, consumer trust will increase. If you are taking extra steps in your daily process to put on-farm food safety at the forefront, you are demonstrating that you care about the quality of your produce.

GHP

Good Hygiene Practices, or GHPs, form the basis of all food hygiene systems that support the production of safe and suitable food. Food business operators must be aware of hazards that could affect their food and ensure that such hazards are properly managed to protect consumer health.

Good Hygiene Practices (GHPs): Fundamental measures and conditions applied at any step within the food chain to provide safe and suitable food. Good Hygiene Practices are the set of requirements to prevent contamination of food in order to provide safe food to the consumers.

GENERAL PRINCIPLES OF FOOD HYGIENE-GHP



GLP(Good Laboratory Practices)

In the food industry, GLP is a quality management system used to control color additives, food additives, food contamination limits, and food packaging

All premises need to be kept clean and tidy

- Clean to remove soil and dirt and sanitise to kill bacteria.
- Cleaning procedures should be available describing how to clean the equipment and the structure of the facility.
- Cleaning chemicals should be stored away from work areas and from food and packaging storage areas.
- Record all cleaning activities in the cleaning report.

Keep surfaces in good condition so they are easy to clean

- Fix broken tiles
- Avoid wooden shelves
- Repair damaged junction seals
- Repair and eliminate rust on any surface

Cleaning will ensure

- Cross contamination is reduced and minimised
- Bacteria does not grow on the surfaces
- Quality of the product or process is of high standard

BAP

The Best Aquaculture Practices is the only third-party aquaculture certification program that encompasses the entire production chain, including the processing plant, farm, hatchery and feed mill.

Best Aquaculture Practices (BAP) is a seafood-specific certification program that addresses the four key areas of responsible seafood—environmental, social, food safety, and animal health & welfare—at each step of the aquaculture production chain

BAP (Best Aquaculture Practices) is one of the most comprehensive and best-known certification systems for ensuring the sustainability of aquaculture products. Four-star certification indicates that the entire system—from feed mill to hatchery, the farm to processing plant—comply with BAP's standards.

Overview of food preservation methods and their underlying principles including novel and emerging methods/principles:

Q: Explain with an examples the various methods for food preservations

A:

Food preservation refers to the different techniques that are applied to food to prevent it from spoiling. The science behind food preservation involves either:

- The destruction of microorganisms responsible for causing food spoilage
- Reducing/eliminating the water (moisture) content from food
- Altering the temperature and other conditions that favor the growth of food microorganisms and thereby retarding microbial growth and replication (thus delaying food spoilage).

Simple household food preservation techniques

1. **Drying :** A number of foods (fruits, vegetables, tubers-cassava, and potatoes) which cannot be stored for long in their fresh state without spoiling can be preserved by drying. Before drying, there should be enough sunshine and foods should be sliced in small pieces for them to dry faster. Dried fruits can be eaten in their dry state (e.g. dried jackfruit), vegetables and potatoes need to be cooked by boiling in water while dried cassava can be ground into flour and used later.
2. **Smoking :** Smoking meat and fish is a highly recommended method for prolonging their storage life. The fish is first cooked over a high fire and then smoke-dried in one to five days (and nights) over a low fire. Fresh-dried fish keeps for up to a week, while hard-dried fish (keeping fish in salt for several weeks) keeps for several months.
3. **Salting:** Salting is a simple food preservation method that can be used to prolong the shelf life of food for a few days. When added to foods, salt takes out moisture and retards microbial growth and replication.
4. **Boiling :** Boiling of foods kills food microbes. Perishable foods can be boiled, cooled and kept in clean containers and then used within a day

Overview of food packaging methods and principles including novel packaging methods:

Q: Describe three levels of packaging the food products

A:

Packaging means the wrapping or bottling of products to make them safe from damages during transportation and storage. It keeps a product safe and marketable and helps in identifying, describing, and promoting the product. “Packing is the preparation of a product or commodity for proper storage and/or transportation.”

Packaging FACTORS considered:

- 1960s convenience
- 1970s light weight
- 1980s source reduction
- From 1990s environmental impact

Primary packaging: Primary packaging is the packaging that contains the product. All of the packaging which surrounds the product when the consumer takes it home. For a multipack of crisps, for example, the primary packaging will be the individual bags and the large bag into which the separate packs are packed.

Secondary packaging: Secondary packaging includes boxes or containers containing specific quantities of primary packages. It is used to group packs together for ease of handling. In the example above of the crisps, several multipacks are packed into printed corrugated cases.

The case is the secondary packaging. Other examples of secondary packaging are shrinkwrap film, and the corrugated board and thermoformed plastic trays used for shelf-ready packaging.

Tertiary packaging: Tertiary packaging includes pallets and large shipping containers for storing and warehousing. It is used to collate secondary packs for ease of transport. One of the most common forms of tertiary packaging is the pallet, along with stretch wrap film and a label, to secure the secondary packs to the pallet and provide a ready means of identification. Roll cages and crates are also examples of tertiary packaging.

Q: Explain the effects of poor packaging on food products, add a note on smart packaging.

A:

Badly designed packs are more likely to:

- increases the risk of shrinkage.
- allow their contents to become damaged in transit

- cause contamination through leakage
- experience product spoilage or damage
- contribute to product loss through misidentification
- lose of contents
- become too ugly to sell
- suffer decreased shelf life and consequent reduced sales

Some consequences of poor packing:

- **Spoilage:** products that have reached their expiry date or gone beyond agreed temperature parameters and are no longer safe to sell to consumers or staff.
- **Damages:** refers to products that have been damaged during the journey to the shelf and in the general store environment.
- **Data errors:** errors in the recording of product details on company systems.
- **Pricing errors:** losses caused by errors in the way in which goods are priced and sold in the business.
- **Delivery errors:** losses caused by incorrect quantities booked to the store inventory but not physically delivered or transferred.
- **Scanning Errors:** errors occurring at the point of sale leading to a discrepancy in the store book stock.

Smart Food Packaging

- Retain integrity and actively prevent food spoilage (shelf-life)
- Enhance product attributes like look, taste, flavor, aroma etc
- Respond actively to changes in product or package environment
- Communicate product information, product history or condition to user
- Assist with opening and indicate seal Integrity
- Confirm product authenticity and act to counter theft.

Ex: film membrane sensor – can check food freshness without opening, detects bacterial growth through pH, is safe in direct contact with food, easily integrated in food packaging.

Q: Classify food packaging methods, and add and explanatory note on the novel food packaging methods

A:

Most sophisticated consumers, Increased working women, Single parent, dual income household, Lifestyle change, Health conscious, Aging population, Educated and knowledgeable consumers, Nutrition, food safety

Food packaging methods:

1. **Tech enabled solutions:** Fibre based materials robotics , Smart packaging , 3D printing, Emotional engagement, Vintage inspired designs, Transparent and clear

labeling, Increased portability, Personalization, Minimal designs , Gradient coloring, Sustainability

2. **Creative food packaging :** soft drinks, seltzer packaging, butter, burger, pizza, cereal, salad, nuts , dried fruit, oxygen scavengers, co2 generators and scavengers, Ethylene scavengers

Quiz:

1. Full form of BAP: **Best Aquaculture Practices**

2. Discuss advantages of preservation of food.it
    - **Safety : Preserving food makes it safe to eat by preventing the growth of germs and microorganisms. This can help prevent food poisoning and foodborne illnesses.**
    - **Quality: Preservation can help maintain the nutritional and sensory quality of food. For example, chemical preservatives can help prevent nutrient loss, and radiation can extend the life of strawberries and mushrooms.**
    - **Availability: Preservation can ensure food is available in remote areas and during the offseason. For example, you can enjoy seasonal fruits like strawberries and mangoes even when they aren't in season**
    - **Sustainability: Preservation can reduce food waste and save resources like water, energy, and land that are used in food production. It can also reduce waste in landfills and your environmental footprint.**
3. What is the most dangerous temperature for food preservation?
4. What is GHP? Explain its importance.

    Good Hygiene Practices are the set of requirements to prevent contamination of food in order to provide safe food to the consumers.They form the basis of all food hygiene systems that support the production of safe and suitable food. Food business operators must be aware of hazards that could affect their food and ensure that such hazards are properly managed to protect consumer health.

5. People, premises, processes, products and procedures is related to **GMP**
6. How do you sterilize food? Give examples.

    Foods that have a pH of more than 4.6, such as meat and most vegetables, must undergo severe heating conditions to destroy all pathogens. These foods are heated under pressure to 121°C for varying times.

    Ex:

    - **All canned foods are sterilized in a retort (a large pressure cooker).**
    - **Severe conditions are applied to ensure that Clostridium botulinum spores are destroyed during processing.The spores are destroyed by heat or are**

inhibited at pH values of less than 4.6. Therefore, a food with a pH of less than 4.6 that is packaged anaerobically.

7. What are the unit operations in food processing?